MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCCMA201 – Digital Tools for Business and Research

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

(Answer all questions, each question carries 2 marks.)

1. Mention any two uses of data analytics in business. [K] [CO1]
2. List the three main views in SPSS/PSPP. [K] [CO1]
3. Give any two summary measures used to describe data numerically. [U] [CO2]
4. Cross tab is a way to bring out the relationship between variables. Justify the statement. [U] [CO2]
5. Differentiate between a paired sample t-test and an independent sample t-test. [U] [CO3]

[5 x 2 = 10]

Part B

Short Answer Questions

*(Answer **5** out of 7 questions, each question carries **5** marks.)*

6. Explain the importance of information in business decision-making. [K] [CO1]
7. Explain the steps involved in creating a data file and defining variables in SPSS/PSPP. [U] [CO1]
8. Explain the process of computing a new variable in SPSS/PSPP with an example. [U] [CO2]
9. Describe the purpose and use of scatter diagrams, P-P plots, and Q-Q plots. [K] [CO2]
10. Explain the procedure for conducting Explore command and interpreting the results. [U] [CO3]
11. A researcher wants to study the effect of two factors (e.g., teaching method and gender) on student performance. Explain how two-way ANOVA can be applied, including assumptions and steps. [A] [CO3]

12. Describe when and why nonparametric tests like Chi-square and Kruskal-Wallis are used. Explain with an example. [A] [CO3]

[5 x 5 = 25]

Part C

Essay Questions

*(Answer **1** out of 2 questions, each question carries **15** marks.)*

13. Illustrate the methods of presenting data graphically in SPSS with examples. [U] [CO2]
14. Explain correlation and regression analysis with assumptions and the step-by-step procedure to analyze the relationship between two variables in SPSS/PSPP. [U] [CO3]

[1 x 15 = 15]